

IP Addressing

◆ What is an IP Address?

An **IP address (Internet Protocol address)** is like a **home address for a device on a network**.

- It's a **unique identifier** that lets devices find and talk to each other.
- Without it, your laptop, phone, or printer wouldn't know where to send or receive data.

✓ **In short:**

An **IP address is a digital address** that identifies a device in a network, just like your home address identifies where you live in the real world.

◇ Types of IP Addresses

◆ IPv4 (Internet Protocol version 4)

- **Address Format:** 32-bit number (e.g., 192.168.1.1).
 - **Range:** About 4.3 billion unique addresses.
 - **Notation:** Written in **dotted-decimal** format (four octets separated by dots).
 - **Example:** 8.8.8.8 (Google's Public DNS server).
 - **Limitation:** Address exhaustion (we've almost run out of IPv4 addresses).
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◆ IPv6 (Internet Protocol version 6)

- **Address Format:** 128-bit number (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334).
 - **Range:** Around 340 undecillion (3.4×10^{38}) addresses – practically unlimited.
 - **Notation:** Written in **hexadecimal** with colons (:).
 - **Example:** 2001:4860:4860::8888 (Google's Public DNS in IPv6).
 - **Advantages:**
 - No shortage of addresses.
 - Built-in **security features (IPSec)**.
 - Better **efficiency** for routing and packet handling.
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◆ Public vs. Private IPs

- **Private IP:** Used inside your home/office network. (e.g., 192.168.0.10, 10.0.0.5). These are **not routable on the internet**.
- **Public IP:** Unique across the entire internet (e.g., 142.250.183.206 → belongs to Google).

◇ Who Assigns Public IP Addresses?

1. **IANA (Internet Assigned Numbers Authority)**
 - Manages global IP address allocation.
 2. **RIRs (Regional Internet Registries)** – distribute IP blocks:
 - **ARIN** (North America)
 - **RIPE NCC** (Europe, Middle East, Central Asia)
 - **APNIC** (Asia-Pacific)
 - **LACNIC** (Latin America, Caribbean)
 - **AFRINIC** (Africa)
 3. **ISPs (Internet Service Providers)**
 - Your ISP leases you a public IP from its assigned block.
 - Example: If you're in Canada using **Bell or Rogers**, they give you a dynamic public IP when you connect your modem.
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◆ Real-Life Examples

- **Home Internet:**
 - You connect your Wi-Fi at home. Your ISP (say **Bell Canada**) assigns you a **public IPv4** like 99.229.45.12.
 - Inside your house, your router gives your devices **private IPs** like 192.168.0.5.
 - **Company Data Center:**
 - A bank in Toronto needs static public IPs for its servers. It requests a block of IPv4/IPv6 addresses from its ISP or directly from **ARIN**.
 - **Google DNS:**
 - IPv4: 8.8.8.8
 - IPv6: 2001:4860:4860::8888
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✓ In short:

- **IPv4** = older, limited, still widely used.
- **IPv6** = newer, massive address space, slowly being adopted.
- **Public IPs** are assigned by **IANA** → **RIRs** → **ISPs** → **You**.

◇ How IP addresses are assigned to Router

- A **router** usually has **two interfaces**:
 1. **WAN (Wide Area Network) side** → This connects to the ISP. It is usually assigned a **public IP** (e.g., 99.229.45.12) by your Internet Service Provider.
 2. **LAN (Local Area Network) side** → This connects to your internal devices (PCs, phones, printers). The router assigns itself a **private IP** (e.g., 192.168.0.1 or 10.0.0.1) to act as the **gateway** for your network.
- Devices in your home (like your laptop) get private IPs (e.g., 192.168.0.10) via **DHCP** from the router.

- The router then uses **NAT (Network Address Translation)** to translate all your devices' private IPs into its single **public IP** when communicating with the internet.
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◆ Example in Real Life

- You have **Bell Canada** internet:
 - Router WAN side: 142.250.183.206 (public IP given by ISP).
 - Router LAN side: 192.168.0.1 (private IP — your devices use this as the gateway).
 - Your laptop: 192.168.0.5 (private IP assigned by the router).

When your laptop goes to `youtube.com`, the router translates your laptop's private IP → its own public IP → sends it out.

◆ Special Case

- In **large enterprise networks**, sometimes **both WAN and LAN** sides can use private IPs if they're behind another layer of firewalls or NAT.
 - Example: A company network may connect to an ISP using **10.x.x.x** private addressing, and only at the edge firewall does it get mapped to a public IP.
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✅ **So yes:** a router almost always has a **private IP on the LAN side**, and sometimes on the WAN side too (if NAT is used by the ISP).

Let's take it from the basics — **IP Classes**, then move to how you, as a beginner, can **set IP scope and assign them** in a real-world scenario.

◆ IP Address Classes (IPv4)

IPv4 addresses are divided into **classes** to define how many **networks** and **hosts** they can support.

1. Class A

- Range: 1.0.0.0 → 126.255.255.255
- Default Subnet Mask: 255.0.0.0 (/8)
- Number of Hosts: ~16 million per network
- Example: 10.0.0.1 (often used in private networks)

2. Class B

- Range: 128.0.0.0 → 191.255.255.255
- Default Subnet Mask: 255.255.0.0 (/16)
- Number of Hosts: ~65,000 per network
- Example: 172.16.0.1

3. Class C

- Range: 192.0.0.0 → 223.255.255.255
- Default Subnet Mask: 255.255.255.0 (/24)
- Number of Hosts: 254 per network
- Example: 192.168.1.1

4. Class D (Multicast)

- Range: 224.0.0.0 → 239.255.255.255
- Used for streaming, multicast groups.
- Example: IPTV broadcasts.

5. Class E (Experimental / Research)

- Range: 240.0.0.0 → 255.255.255.255
- Rarely used in real life.

◆ Reserved Private IP Ranges

For internal networks (non-routable on internet):

- Class A: 10.0.0.0 → 10.255.255.255
- Class B: 172.16.0.0 → 172.31.255.255
- Class C: 192.168.0.0 → 192.168.255.255

◇ Setting IP Scope

👉 “Scope” means deciding the **range of IPs** you will allow in your local network.

Example: Home Network

- Suppose your router LAN IP is 192.168.1.1.
- You can set a **DHCP Scope** on the router:
 - Start IP: 192.168.1.100
 - End IP: 192.168.1.200
 - Subnet Mask: 255.255.255.0
 - Gateway: 192.168.1.1
 - DNS: 8.8.8.8 (Google DNS)

This means your devices (laptop, phone, printer) will automatically get IPs in the range **192.168.1.100** → **192.168.1.200**.

◆ How to Assign IPs in Real Life

1. Automatic Assignment (DHCP – Most Common)

- The router automatically gives IPs from the defined **scope**.
 - Example: You connect your phone → router assigns it 192.168.1.105.
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2. Manual Assignment (Static IPs)

- You go to your device's network settings and set:
 - IP Address: 192.168.1.50
 - Subnet Mask: 255.255.255.0
 - Gateway: 192.168.1.1
 - DNS: 8.8.8.8

Used when:

- You want a server (like a file server or printer) to always keep the same IP.
 - Example: Assign your printer 192.168.1.250.
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◆ Real-Life Example (Step by Step)

Scenario: Setting up Wi-Fi at home

1. ISP gives router a **public IP** (say 99.229.45.12).
 2. Router LAN IP = 192.168.1.1 (**Private**).
 3. Set **DHCP scope**: 192.168.1.100 – 192.168.1.200.
 4. Laptop connects → gets 192.168.1.101.
 5. Printer → set static IP = 192.168.1.250.
 6. All devices use **Gateway (192.168.1.1)** to reach the internet.
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✓ Summary for you as a novice:

- Use **Class C** (192.168.x.x) for small home/office networks.
 - Let your router's **DHCP** handle most devices.
 - Assign **static IPs** only to special devices (like servers, printers, NAS).
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◇ Subnet Mask

A **subnet mask** defines **which part of an IP address is the "network" portion and which is the "host" portion**.

- It's used to figure out **which devices are on the same local network**.

Example

- IP Address: 192.168.1.10
- Subnet Mask: 255.255.255.0

👉 Here's what it means:

- 255.255.255 → Fixed (network portion)
- .0 → Flexible (host portion, devices can vary from 1 to 254)

So, devices from **192.168.1.1 → 192.168.1.254** are on the **same network**.

✅ If your computer has 192.168.1.10 and your printer has 192.168.1.50, they can talk **directly** because they're in the same subnet.

◇ Gateway

A **default gateway** is the **device (usually your router)** that connects your local network to the **outside world (internet)**.

- If two devices are in the same subnet → they talk directly.
- If a device wants to talk **outside its subnet (like a website)** → it sends the request to the **gateway**.

Example

- Laptop IP: 192.168.1.10
- Subnet Mask: 255.255.255.0
- Gateway: 192.168.1.1 (router's IP)

👉 When you go to `youtube.com`:

1. Your laptop sees YouTube's server isn't in 192.168.1.x.
 2. It sends the request to 192.168.1.1 (gateway).
 3. The router forwards it to the internet using your **public IP**.
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◆ Real-Life Analogy

- **Subnet mask** = defines your "neighborhood" (who lives on the same street as you).
 - **Gateway** = the "exit door" from your neighborhood to reach other neighborhoods (internet).
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◆ Quick Example Setup

- **Router LAN IP (Gateway):** 192.168.1.1
- **Subnet Mask:** 255.255.255.0
- **Laptop IP:** 192.168.1.10
- **Phone IP:** 192.168.1.20

👉 Both devices can chat directly (same subnet). When they need internet, they go through 192.168.1.1 (gateway).

✅ In short:

- **Subnet mask** tells devices who's "local."
- **Gateway** is the router that connects your local network to everything else.

Static vs Dynamic IP addresses.

◆ Static IP Address

A **static IP** is an IP address that **doesn't change**. It is manually assigned to a device and stays the same until you change it.

✅ Characteristics

- Permanent (fixed address).
- Easier to find a device on the network.
- Often used for servers, printers, or security cameras.

📌 Example

- Your **office server** might always be 192.168.1.50.
- Your **company website server** could have a public static IP: 142.250.190.14.

👉 If you type that IP in your browser, you'll always reach the same server.

◆ Dynamic IP Address

A **dynamic IP** is **automatically assigned** by a **DHCP server** (usually your router or ISP). It can change whenever you reconnect or after a set period.

✓ Characteristics

- Temporary (changes over time).
- Saves IP address space.
- More secure for home users (harder for hackers to target).

📌 Example

- At home, your ISP gives your router a **dynamic public IP**. Today it could be 99.229.45.12, tomorrow 99.229.45.89.
 - Your laptop inside Wi-Fi might get 192.168.1.101 today, but if you disconnect/reconnect, it could become 192.168.1.110.
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◆ Real-Life Analogy

- **Static IP** = Owning a house with a permanent address (easy for friends to find you anytime).
- **Dynamic IP** = Living in a hotel where the room number changes each time you check in (your friends call the reception first).

When to Use Each

- **Static IP**
 - Hosting a website.
 - Running a VPN server.
 - Setting up a network printer.
 - Remote access to CCTV.
- **Dynamic IP**
 - Normal home internet use (browsing, streaming, gaming).
 - Everyday laptops, phones, tablets.

✓ In short:

- **Static IP** = fixed, best for servers.
- **Dynamic IP** = changes, best for regular users.

◇ What is APIPA?

APIPA stands for **Automatic Private IP Addressing**.

- It is a **fallback mechanism** used by Windows (and some other OSes) when a device **can't get an IP address from a DHCP server**.

- The system automatically assigns itself an IP in a special reserved range.

◆ APIPA Range

- **Range:** 169.254.0.1 → 169.254.255.254
- **Subnet Mask:** 255.255.0.0
- **Gateway:** None (since it can't reach outside the local network).

◆ When Does It Happen?

1. Your device is set to “Obtain IP Automatically” (DHCP).
2. The **DHCP server is unavailable** (router down, cable unplugged, server crash).
3. Windows assigns an APIPA address so the device can still talk to **other devices in the same subnet**, but **not the internet**.

◆ Real-Life Example

- You connect your laptop to Wi-Fi, but the router's DHCP server is broken.
- Your laptop tries to get an IP... fails... so Windows assigns:
 - IP Address: 169.254.23.45
 - Subnet Mask: 255.255.0.0
 - Gateway: (blank)

👉 You can **share files or print** with another device on the same APIPA range, but you **can't access the internet**.

How to Recognize It

- Run `ipconfig` on Windows.
- If you see an address starting with 169.254, it means **DHCP failed → APIPA assigned**.

◆ Analogy

Think of APIPA like this:

- You walk into a hotel (network) expecting the receptionist (DHCP server) to give you a room key (IP).
- The receptionist is missing.
- So you pick a **temporary room** in a “default” hallway (169.254.x.x) — but you can't leave the hotel (no internet).

✅ In short:

APIPA = A self-assigned IP address in the 169.254.x.x range when DHCP fails. Good for **local communication**, but **no internet access**.